

Black-box Active Learning for Discovering New Explanations



Derivative-Free Optimization Symposium

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Paul SAVES, Nicolas VERSTAEVEL, Benoît GAUDOU



Context

- *Multi-agent systems & simulation*
- *Explainable AI for complex systems*
- *Socio-environmental digital twins*

600 lab members
Yearly budget 5.1M€ (2023)



Membre de l'Université Toulouse
Capitole

Created in 1229
3 campus
750 Teaching staff
740 administrative staff
21000 students
151 national diplomas

ANR-JCJC MIMICO: MIMIcking the COmplexity of agent-based models

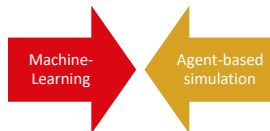
■ Team



Nicolas Verstaavel
Principal Investigator



Paul Saves
Postdoctoral fellow



■ Extended team



Manon Prédhumeau



Benoit Gaudou



Frederic Amblard



Jordan Levy



Clément Blanco-Volle



Matthieu Mastio



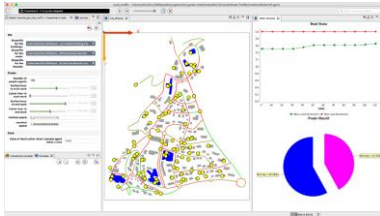
Complex Systems Simulations

■ Agent-based simulations

- Useful to study **complex socio-environmental** challenges (climate change, water scarcity, urbanization, pollution, energy transition, sustainability...)
- Simulation models can be abstracted as a program that transforms a set of **inputs** into a set of **outputs**.
- **Calibration/Exploration/Optimization:** goal-oriented and user-defined automated exploration of a model's outcomes to study a given emerging and decentralized phenomenon



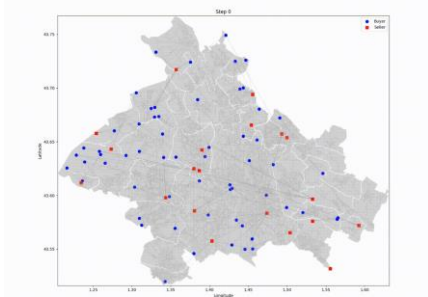
<https://gama-platform.org/>



From Agent based simulations to emergent behaviors

- Industrial Symbiosis model - MASTIO

- Initial **random** placement, **RL algo.**, decreasing temp.

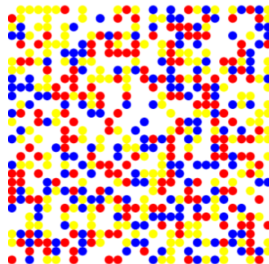


- 5 cont.: product scarcity, firm density, transport. cost, disposal cost, cluster spread



- Schelling segregation model

- Initial **random** placement, **random** move if unsatisfied.



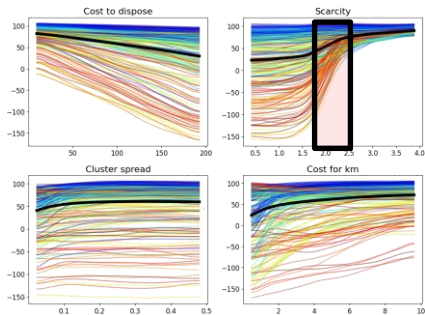
- 3 integers: grid size, number of socio-types, agent perception distance & 2 cont.: intolerance of uninhabitant and population density.

<https://mobiliscope.cnrs.fr/en/geoviz/montreal>

Motivation: Tipping Points in Dual Spaces

Parameter Space (MASTIO ICE/PDP)

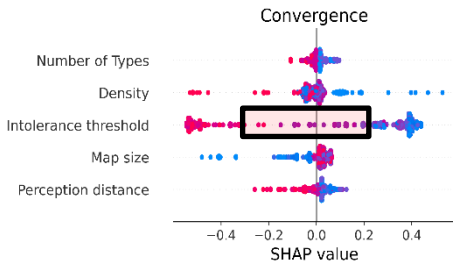
Regression



- Tipping point: Prediction collapses/diverges at specific parameter thresholds.
- Transient phases may exhibit scale-free (power-law) dynamics.

Explanatory Space (Schelling SHAP)

Classification



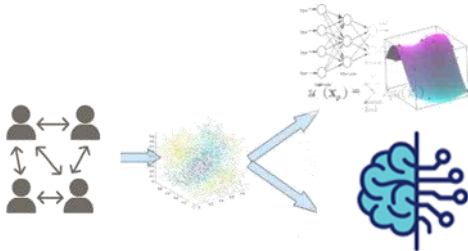
- Internal logic shift: Appearance of isolated "islands" where features suddenly dominate convergence.

RQ: Can hunting for explanatory novelty help find physical tipping points?

Limits of Complex Systems Simulations

- **High computation costs:** large numbers of agents inter. **over time and space**. *High-dimensional mapping via LHS is intractable.*
- **Opacity of outcomes:** intricate interactions among agents often lead to results that are **hard to trace back** to specific inputs or rules. *Not easily decomposable.*
- **Black-Box:** No gradients to guide the search towards interesting behaviors.
- **Needle in a Haystack:** Traditional Active Learning minimizes *prediction uncertainty*, wasting simulations in noisy but homogeneous regions while missing critical tipping points.

Explainable AI for Surrogate models



- **Mimic** a complex system, black box $\Rightarrow$ ease calibration, scenarios exploration, and optimization

- Types of models used $\Rightarrow$ Models are **complementary: complexity or explicability** based on **machine learning**

Pipeline & Innovations

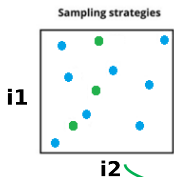
Active Learning and localities

Local explainability

Q2: What does SM tell us about the parameter space?

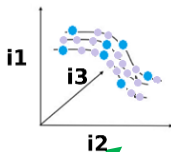
Exploration + Calibration

Initial
dataset



Refined
dataset

Surrogate model



Q1: How to efficiently train SM from a few samples?

Initial + Refined datasets

Global explainability

Q3: How does the SM inform us about the model dynamics?

Analysis + XAI

Augmented
dataset



The Core Concept: Explanatory Diversity

- Instead of looking only at the prediction y , we extract feature contribution profiles using XAI

SHAP values attribute, for a given prediction, how much each feature contributes to the deviation between that prediction and the model's average prediction, by fairly distributing that difference across features.

$$\phi_{i,j} = \sum_{U \subseteq [1:m] \setminus \{i,j\}} \frac{|U|! (m - |U| - 2)!}{(m - 1)!} (v(U \cup \{i, j\}) - v(U \cup \{i\}) - v(U \cup \{j\}) + v(U)),$$

- **Goal:** Explicitly sample regions where the explanation of the model fundamentally changes.

Min-Max Cosine Proxy

From Slow KDE to Fast Cosine Proxy:

- My initial idea relied on high-dimensional Kernel Density Estimation (KDE) on SHAP values:

$$\rho(S(x)) = \frac{1}{N} \sum_{i=1}^N K_h(S(x) - S(x_i))$$

KDE suffers from the curse of dimensionality, decouples parameters, creates normalization issues, and is computationally prohibitive.

The Solution: We implemented a **Fast Density Proxy via Cosine Distance**. High cosine similarity (CS) to the nearest neighbor translates directly to high explanatory novelty.

The direction of the XAI change, even when compared to the most similar explanation!

→ **Max-CS to the Nearest Neighbor (1-NN) in SHAP space** ~ “Max-Surprising”

Dual-Surrogates & SHAP Data Augmentation

Bypassing the ABM Computational Bottleneck:

- **Primary GP (Physical Space):** Trained directly on the true ABS outputs. It acts as an ultra-fast proxy for the simulation (*Dataset $\mathcal{D}$*).
- **Data Augmentation Trick:** ExactSHAP computations are exponentially expensive $\mathcal{O}(2^M)$. We use the Primary GP to generate thousands of cheap SHAP profiles (*Low-Fidelity XAI data*).
- **The Result:** A massive, augmented dataset combining a few exact ABM profiles (*High-Fidelity XAI data*) and thousands of GP-derived profiles, providing dense explanatory context evaluated to give *Low-Fidelity and High-Fidelity XAI Novelty*.
- **Secondary Multi-fi GP (XAI Space):** Trained on the 2-fidelities *XAI Novelty* sets.

Bi-Objective Optimization

Framing Dynamic-US as a Bi-Objective Problem:

Rather than a static heuristic, the acquisition strategy sweeps a Pareto front to simultaneously:

- 1) **Maximize Explanatory Exploitation:** The MFK provides both a novelty prediction and its uncertainty, allowing us to compute EI_{MFK} (the potential for discovering a highly novel explanation).
- 2) **Maximize Physical Exploration:** Predictive variance in input space Var_{GP} .

Dynamic Scalarization: We dynamically sweep a Pareto front, starting with spatial uncertainty ($\alpha_{start} = 0$) and explanatory novelty ($\alpha_{end} = 1$)

$$O_{Dynamic}(x) = (1 - \alpha) Var_{GP}(x) + \alpha EI_{MFK}(S(x))$$

Pseudocode

Algorithm 1: Black-box Active Learning for Discovering New Explanations

Input: Expensive black-box simulator f , Feature design space $\mathcal{X}$, Initial budget n_{doe} , Total budget N ,
Low-fidelity explanation budget n_{expl_lowfi}

Output: Explored dataset $\mathcal{D} = \{x_{doe}, y_{doe}\}$

// Step 1: Initial Dataset Generation

- 1 Sample n_{doe} points x_{doe} in $\mathcal{X}$ using Latin Hypercube Sampling (LHS);
- 2 Evaluate black-box simulator: $y_{doe} \leftarrow f(x_{doe})$;
- 3 Initialize dataset $\mathcal{D} \leftarrow \{x_{doe}, y_{doe}\}$;
- 4 Initialize dynamic scalarization weight: $\alpha^{(0)} \leftarrow \alpha_{start}$;
- 5 Define step size $\Delta\alpha = \frac{\alpha_{end} - \alpha_{start}}{N - n_{doe} - 1}$;
- 6 Initialize iteration counter $t \leftarrow 0$;

Pseudocode

```
// Active Learning Loop
7 while  $|\mathcal{D}| < N$  do
    // Step 2: Explanation Inputs and Surrogate Training
    8 Train a noisy Gaussian Process (GP) on the current dataset  $\mathcal{D}$ ;
    9 Sample  $n_{xpl\_lowfi}$  points  $x_{xpl\_lowfi}$  in  $\mathcal{X}$ ;
    10 Compute ExactSHAP values  $S$  for  $(x_{xpl\_lowfi} \cup x_{doe})$  using the GP as a proxy, with  $x_{doe}$  as the background
        reference;

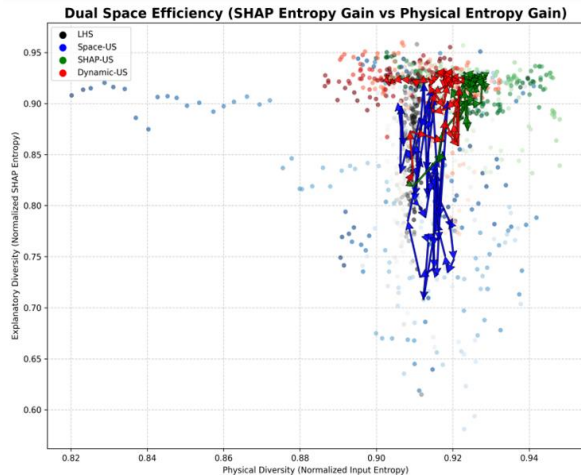
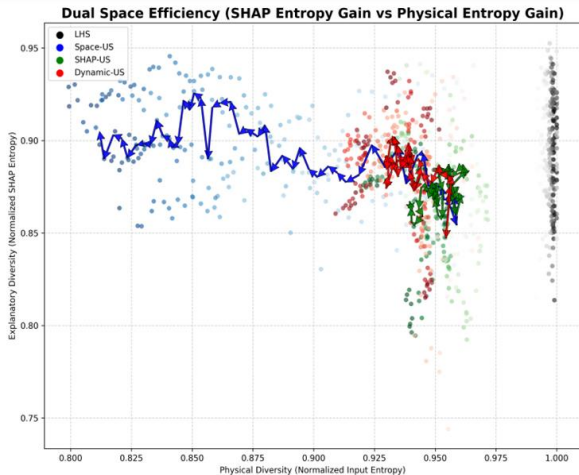
    // Step 3: Explanation Novelty
    11 Compute proxy novelty  $y_{nov\_total}$  on the SHAP values  $S$  using Cosine Distance (k-NN);
    12 Extract high-fidelity novelties:  $y_{nov\_high} \leftarrow y_{nov\_total}(x_{doe})$ ;
    13 Extract low-fidelity novelties:  $y_{nov\_low} \leftarrow y_{nov\_total}(x_{xpl\_lowfi})$ ;
    14 Train a Multi-Fidelity Gaussian Process (MFGP) to predict explanation density using:
        • High-fidelity data:  $\{x_{doe}, y_{nov\_high}\}$ 
        • Low-fidelity data:  $\{x_{xpl\_lowfi}, y_{nov\_low}\}$ 

    // Step 4: Dynamic Hybrid Bayesian Optimization
    15 Find the next optimal point  $x_{opt}$  by maximizing the Dynamic Hybrid Acquisition function (balancing GP
        Variance and MFGP Expected Improvement) via Differential Evolution:
        
$$x_{opt} = \arg \max_{x \in \mathcal{X}} \left[ (1 - \alpha^{(t)}) \cdot \tilde{\sigma}_{GP}^2(x) + \alpha^{(t)} \cdot \bar{E}_{MFK}(S(x)) \right];$$


    // Step 5: Update
    16 Evaluate the expensive black-box at the new point:  $y_{opt} \leftarrow f(x_{opt})$ ;
    17 Update inputs:  $x_{doe} \leftarrow x_{doe} \cup \{x_{opt}\}$ ;
    18 Update outputs:  $y_{doe} \leftarrow y_{doe} \cup \{y_{opt}\}$ ;
    19 Update dataset:  $\mathcal{D} \leftarrow \{x_{doe}, y_{doe}\}$ ;
    20 Update scalarization weight:  $\alpha^{(t+1)} \leftarrow \alpha^{(t)} + \Delta\alpha$ ;
    21  $t \leftarrow t + 1$ ;
22 end
```

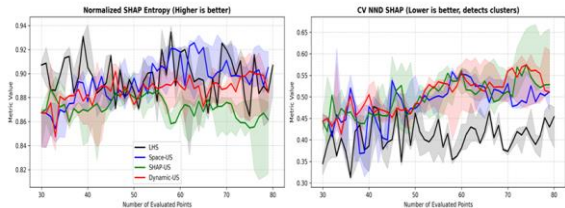
Results & Analysis

Convergence and Stability over Time

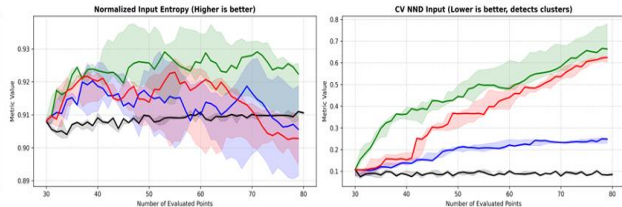
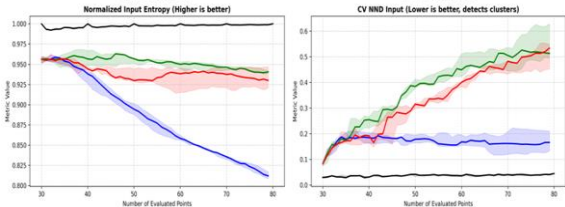
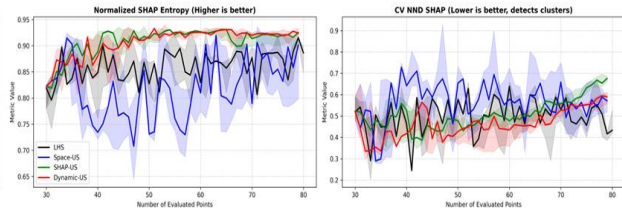


Convergence and Stability over Time

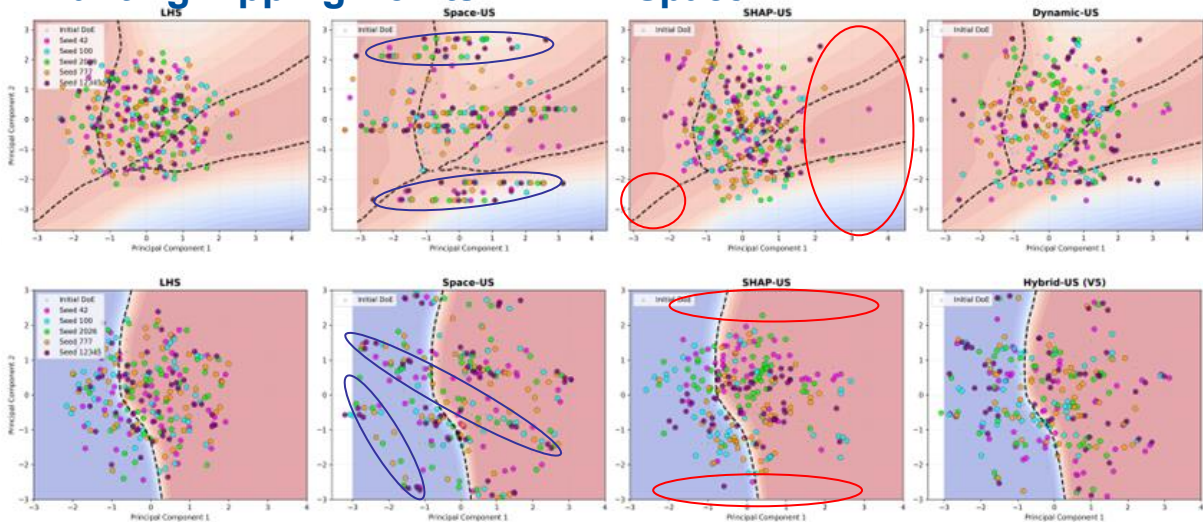
Evolution of Metrics across 50 Iterations (Median & IQR over 5 Seeds)



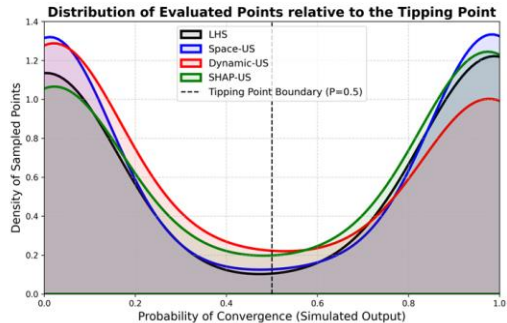
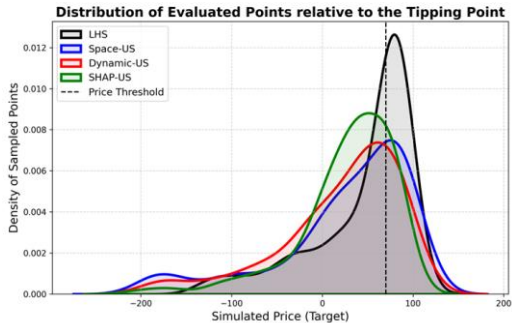
Evolution of Metrics across 50 Iterations (Median & IQR over 5 Seeds)



Hunting Tipping Points in PCA Space

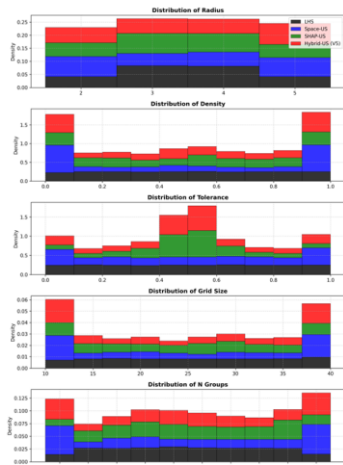


Hunting Tipping Points in the Output Space

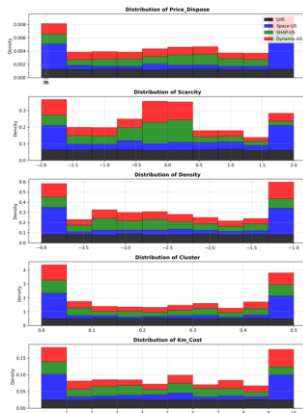


Hunting Tipping Points in the Parameter Space

1D Histograms per Variable (Aggregated over 5 seeds)

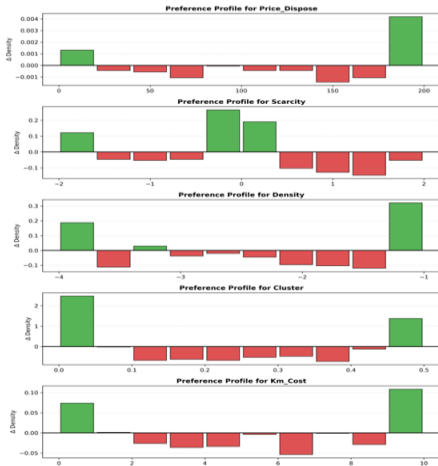


1D Histograms per Variable (Aggregated over 5 seeds)

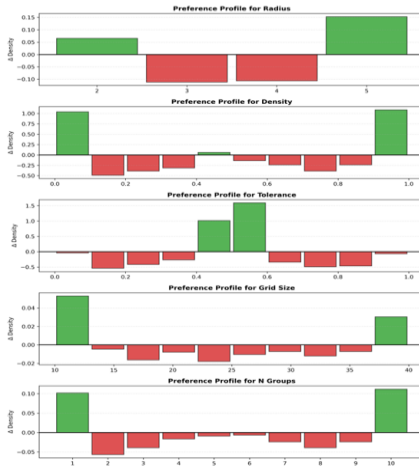


Hunting Tipping Points in the Parameter Space

Targeted Search Profile (Δ Density: Hybrid-US minus LHS)



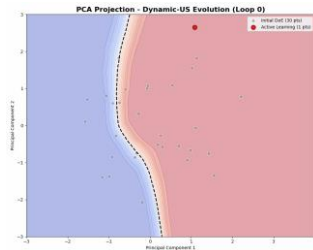
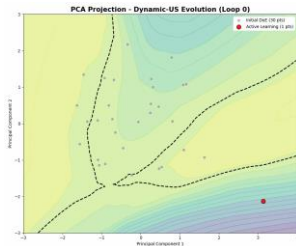
Targeted Search Profile (Δ Density: Hybrid-US minus LHS)



Conclusions & Perspectives

Conclusions

- A model-agnostic, multi-fidelity XAI loop that achieves deep structural understanding rather than just raw predictive accuracy.
- Demonstrates that optimizing for explanatory diversity uncovers tipping points exponentially faster than classic DoE or SUR.
- Validated on robust Agent-Based Models (MASTIO, Schelling).



Perspectives & Future Works

- **Integrated Variance Reduction (IMSE):** Replacing pointwise GP variance with Integrated Mean Squared Error to solve the extrapolation artifacts at the boundaries of the hypercube.
- **Active Subspace Projection:** Linking SHAP tracking to the dynamic rotation of the Active Subspace. This provides a theoretical pathway to scale dimensions by reducing ExactSHAP overhead.
- **Multi-Objective DFO:** Replacing linear scalarization with Expected Hypervolume Improvement to capture non-convex regions of the Pareto front.
- **Trust-Region Bayesian Optimization (TREGO):** Using dynamically sized local trust regions to handle the inherently non-stationary landscapes of SHAP novelty.
- **Global Solvers (CMA-ES):** Upgrading the acquisition optimizer from Differential Evolution to Covariance Matrix Adaptation Evolution Strategy to avoid local minima traps in the multimodal novelty surface.



MIMICO

Thank you for your attention!



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